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Paleomagnetism of the Kvarner islands, Yugoslavia

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ABSTRACT

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Nearly 400 samples were drilled at 44 localities from Cretaceous and Eocene carbonates and Eocene flysch, all in para-autochthonous position.

As a result of thermal cleaning most of the Cretaceous localities yielded statistically well-defined mean directions. Thus overall means were calculated for the Cretaceous of Cres and Krk islands, respectively (Cres, six localities: $D = 308°$, $I = 45°$, $k = 26$, $\alpha_{95} = 13.3°$, $D_c = 330°$, $I_c = 48°$, $k = 33$, $\alpha_{95} = 11.9°$; Krk, seven localities: $D = 342°$, $I = 42°$, $k = 12$, $\alpha_{95} = 17.8°$, $D_c = 348°$, $I_c = 48°$, $k = 26$, $\alpha_{95} = 12.1°$).

These directions, corrected for tilt, agree well with those determined for rocks of similar age from the fold belt north of "autochthonous' Istria, the Trieste Karst and the Southern Alps, while exhibiting clockwise rotation of about 30° with respect to the rigid part of the Adriatic region.

The natural remanent magnetization (NRM) of most of the Eocene sediments was either too weak to be cleaned properly or became erratic before characteristic remanent magnetization (ChRM) could be isolated. The mean directions of the Eocene localities with ChRM differ so much, both before and after tectonic correction, that an overall mean was not calculated.

Despite the scatter, certain trends could be recognized. Counter-clockwise rotation characterizes most of the Eocene sites. The two sites in Krk with clockwise rotation may belong not to the Dalmatian but the High Karst zone.

1. Introduction

Paleomagnetic results for the Cretaceous of the northwestern part of the Dinaric fold belt have indicated significant counter-clockwise rotation with respect to the present north (Márton and Veljovic, 1987). The magnitude of the rotation, however, could not be determined precisely, as the overall mean directions were practically the same before and after tectonic correlation—this often occurs when strata are sub-horizontal and the scatter for each locality is relatively high because of weak characteristic remanent magnetisation (ChRM).

Thus the problem of relative movement of the para-autochthonous with respect to the more rigid 'autochthon' of Istria was not solved.

Sampling was therefore extended to Krk, Rab and Pag islands, where carbonates of Cretaceous age, flysch and Foraminifera limestones of Eocene age were drilled. With the cores studied earlier from Cres, the collection from the Kvarner islands comprises 400 samples drilled at 44 sites.

2. Outline of tectonics

The Kvarner islands belong to the Dalmatian–Herzegovinian tectonic zone of the Dinarides (Dimitrijevic, 1982). They are built of numerous synclines and anticlines.

The tectonic style is not uniform over the whole area. In the Vinodol unit (Fig. 1), narrow, strongly compressed and often overturned anticlines and

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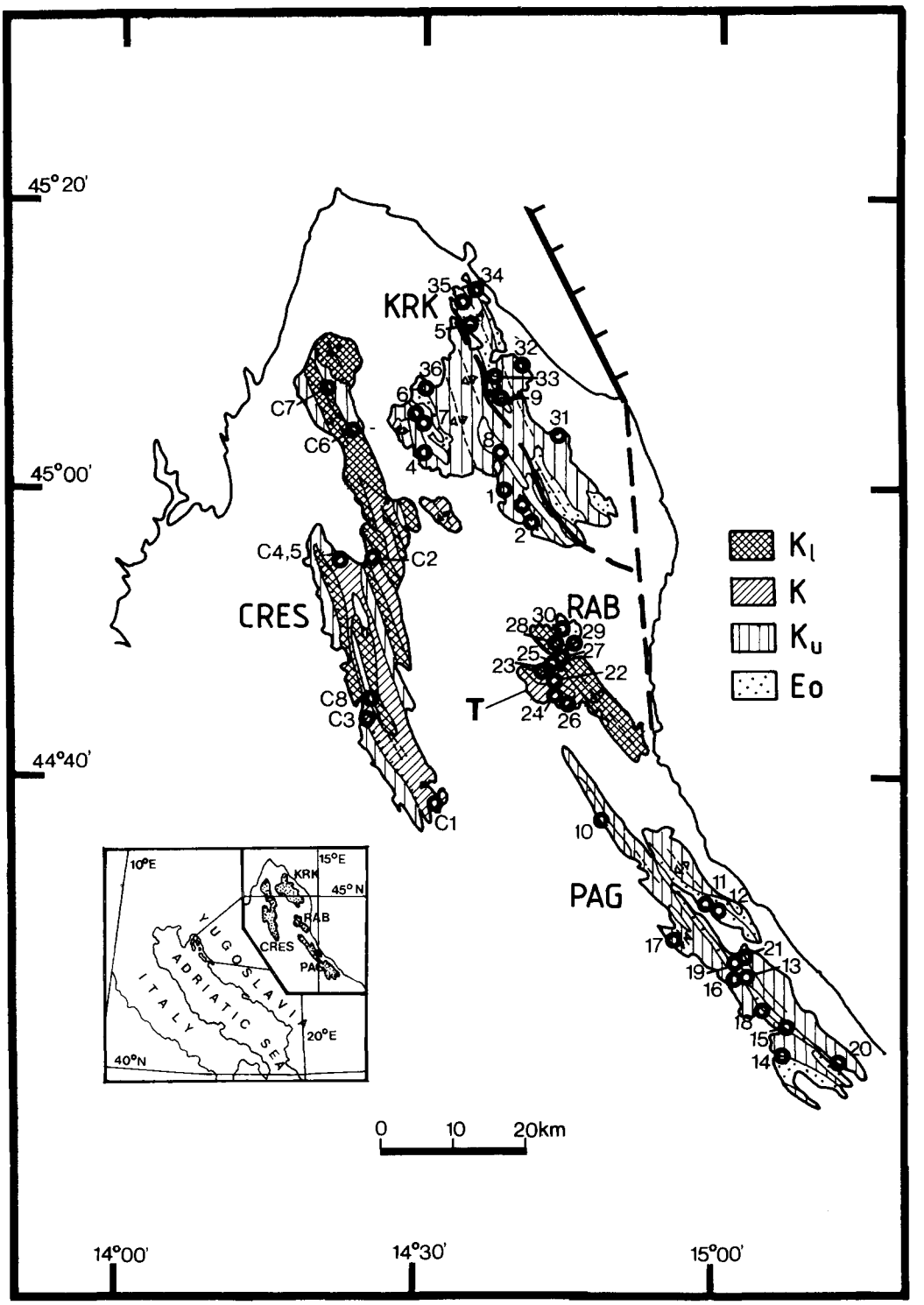


Fig. 1. Geological sketch-map of the Kvarner islands with an index map of the Periadriatic area and with the sampling sites numbered 1–36 for Krk, Rab and Pag islands and C1–C6 for Cres. Kl: early Cretaceous; K: Cretaceous in general; Ku: late Cretaceous; Eo: Eocene. Dominant tectonic features, such as synclines, anticlines and overthrusts, are marked with conventional symbols. The Vinodol unit is bounded by the broken heavy line crossing Krk island, the fault of the Velebit channel in the east and a major overthrust in the NW.

synclines alternate. Open folds are characteristic elsewhere, with a series of overthrusts in Cres island.

The folds are of Dinaric trend (NW–SE) and are fairly uniform in orientation, except in Rab island, where one of the anticlines (marked T in Fig. 1) is oblique to the general structural trend.

The tectonic subdivision of the Kvarner islands is based on paleostructural features, different stratigraphic development and recent structural characteristics. In addition to the already mentioned Vinodol and Cres units, the rest of Krk and Rab island (except anticline T) may be regarded as one unit, whereas Pag island represents an independent unit.

The Eocene is thought to be the epoch when large plicative structures started to form in the northern part of the Dinarides (Dimitrijevic, 1982). Later, during the Styrian orogeny, they were again affected by strong contraction (Grubic, 1980).

3. Sampling and laboratory treatment

Oriented cores were collected from different ages (early–late Cretaceous and Eocene) and lithologies (limestones, dolomites and flysch) at geographically distributed sites (Fig. 1) which represented all tectonic units of the Kvarner islands.

The initial remanences were measured on sister specimens with a JR-3 spinner in Grocka, with JR-4 spinners and on a cryogenic magnetometer in Budapest. One or more specimens per sample were subjected to progressive thermal cleaning in Budapest, after alternating field (AF) cleaning at a peak field of 5 mT. Any minerological change

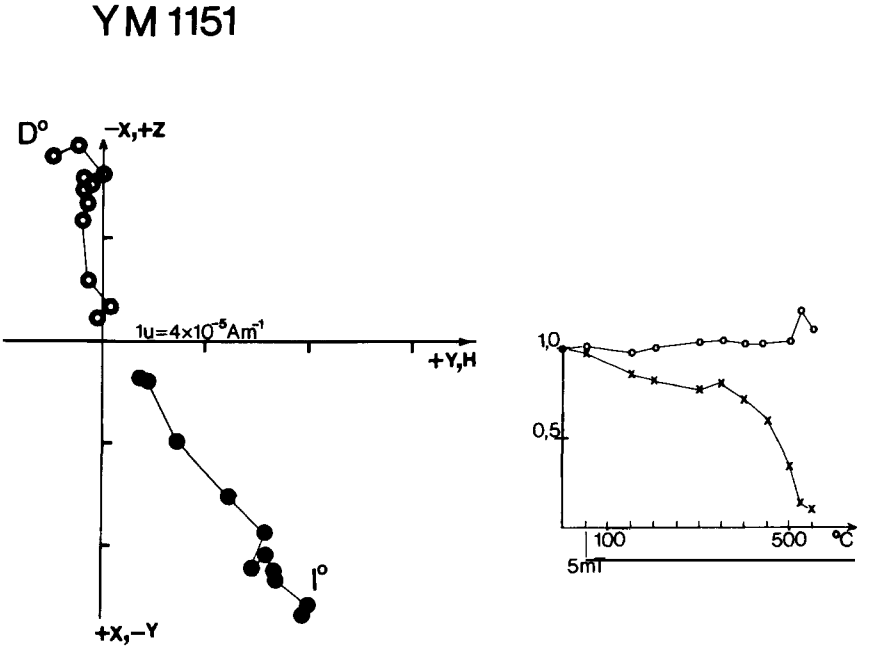


Fig. 2. Typical behaviour of Cretaceous samples during demagnetization. Krk, dolomite, early Cretaceous, site 6. Modified Zijderveld projection (circle: declination; dot: inclination) of the magnetic vector (left) and normalized intensity (cross) and susceptibility (circle) as a function of demagnetizing field and temperature (right). Demagnetization steps on the orthogonal plots correspond to those shown by the plots on the right.

TABLE 1

Summary of paleomagnetic results from Krk, Pag and Rab islands

Sampling site	Rock type age	N/N_0	D D_c	I I_c	k k	α_{95} α_{95}	$R\%$	Cleaning remarks
RAB, Rab anticline (may be continuation of Krk anticline; 44°48′N, 14°45′E)								
29. Lopar, San Marino	Flysch mid–late Eocene	5/12	123 137	−52 −55	8 8	28.3 28.3	100	200–400 °C
30. Lopar	Flysch mid–late Eocene	0/7	–	–	–	–	–	Scatter too high
27. Slanisce	Limestone early–mid Eocene	7/8	183 166	−36 −49	94 94	6.3 6.3	95	300 °C
28. Slanisce	Limestone Tu–Se	0/6	–	–	–	–	–	Too weak
RAB, Topole anticline (44°46′N, 14°42′E)								
25. Kastelina	Limestone mid–late Eocene	7/16	289 321	42 35	16 16	15.8 15.8	0	300–450 °C
22–23. Kampor	Limestone early–mid Eocene	13/17	67 72	−37 −18	73 73	4.9 4.9	100	300–400 °C
24. Kampor	Limestone Ce–Tu	0/6	–	–	–	–	–	Too weak
26. Frkanj	Limestone Ce–Tu	0/6	–	–	–	–	–	Too weak
CRES (44°59′N, 14°24′E)								
C3. Osor	Limestone Ce–Tu	12/12	320 346	53 41	52 52	7.2 7.2	0	100–400 °C
C4. Valun	Limestone Ce–Tu	7/7	293 312	31 58	30 30	11.0 11.0	0	10 mT 150–500 °C
C5. Valun	Limestone Ce–Tu	11/12	306 306	38 38	36 20	7.7 10.6	0	150–500 °C
C6. Predoscica	Limestone Ce–Tu	7/7	292 341	45 46	31 31	10.9 10.9	0	400 °C
C1. Punta Kriza	Limestone Brm–Apt	11/13	322 324	35 47	29 56	8.6 6.1	0	300–500 °C
C2. Krcina	Limestone Brm–Apt	9/9	324 348	63 50	135 135	4.4 4.4	0	150–500 °C
KRK, Omisalj-Vrbnik, anticline (45°12′N, 14°35′E)								
34. Omisalj	Flysch mid Eocene	7/8	57 59 109	−23 2 −6	14 16 12	16.5 15.5 17.7	100	300 °C Normal Overturned
31. Dubni	Limestone Tu	5/12	180 107	−80 −41	17 17	19.1 19.1	100	5 mT–250 °C
32. Silo	Dolomite Tu	5/8	357 6	50 59	50 40	10.9 12.2	20	150 °C
KRK, Omisalj-Baska, syncline (45°10′N, 14°35′E)								
5. Njivice	Limestone mid Eocene	10/10	46 39	6 46	140 140	4.1 4.1	0	450–500 °C
33. Dolovo	Limestone mid Eocene	3/6	171 167	−27 −20	54 54	17.0 17.0	100	400 °C
35. Omisalj	Limestone mid Eocene	2/6	277 284	−15 −45	– –	–	–	5 mT
9. Kras	Limestone Ce–Tu	3/6	324 350	38 53	170 170	9.5 9.5	0	150–200 °C

TABLE 1

Sampling site	Rock type age	N/N_0	D D_c	I I_c	k k	α_{95} α_{95}	$R\%$	Cleaning remarks
KRK, Krk, anticline (45°05′N, 14°35′E)								
2. Stara Baska	Limestone	6/8	319	52	43	10.3	0	5 mT–200 °C
	Tu–Se		337	59	43	10.3		
3. Kuka	Limestone Tu–Se	0/7						Too weak
1. Konobe	Dolomite Ce–Tu	0/9	–	–	–	–	–	Scatter too high
36. Porat	Dolomite	6/7	331	44	19	15.8	0	5 mT
	Ce–Tu		323	33	19	15.8		
4. Pinezic	Limestone Ce–Tu	0/7	–	–	–	–	–	Too weak
8. Sv.Dunat	Limestone	5/8	320	34	35	13.5	0	150–400 °C
	early Cretaceous		348	42	35	13.5		
6. Kukurik	Dolomite	8/8	354	46	46	8.2	0	400–500 °C
	early Cretaceous		345	40	46	8.2		
7. Poljica	Dolomite	5/6	203	−14	28	14.8	0	350–400 °C
	Early Cretaceous		194	−39	28	14.8		
PAG, Saska, anticline (44°30′N, 15°00′E)								
21. Bosane	Flysch mid–late Eocene	0/8	–	–	–	–	–	Great circle distribution 200–375 °C
12. Metajna	Flysch mid Eocene	0/9	–	–	–	–	–	Scatter too high
11. Zubovici	Limestone early–mid Eocene	0/10	–	–	–	–	–	Scatter too high
PAG, Novalja, anticline (44°26′N, 15°03′E)								
10. Melnica	Limestone mid–late Eocene	0/6	–	–	–	–	–	Too weak
13. Zameti	Flysch mid–late Eocene	0/7	–	–	–	–	–	Great circle distribution 150–375 °C
15. Kukovic	Flysch mid–late Eocene	0/8	–	–	–	–	–	Great circle distribution 300–525 °C
19. Pag, Bellevue	Flysch mid–late Eocene	0/9	–	–	–	–	–	Great circle distribution 300–400 °C
20. Miskovici	Flysch mid–late Eocene	0/14	–	–	–	–	–	Great circle distribution 300–400 °C
14. Povljana	Limestone Eocene	0/10	–	–	–	–	–	Scatter too high
17. Mandre	Limestone	5/8	330	46	71	9.1	0	150–250 °C
	Se		16	41	71	9.1		
16. Duboka Draga	Limestone Ce–Tu	0/9	–	–	–	–	–	Too weak
18. Kosljun	Limestone Ce–Tu	0/6	–	–	–	–	–	Too weak

Mean geographical co-ordinates are given in brackets.
N/N_0, numbers of stable/collected samples.
D, I, declination and inclination before tilt correction.
D_c, I_c, declination and inclination after tilt correction.
k, precision parameters.
α_{95}, circle of confidence.
$R\%$, percentage of samples with reversed polarity.
Age abbreviations: Se—Senonian, Tu—Turonian, Ce—Cenomanian, Apt—Aptian, Brm—Barremian.

during heating was monitored by remeasuring the susceptibility before each heating step.

4. Results

Although weak, the natural remanent magnetization (NRM) of the Cretaceous samples was measurable. On cleaning, however, the intensity often dropped very rapidly to the noise level of the magnetometers. For this reason, samples had to be rejected when calculation of the site-means or the definition of ChRMs was impossible, as it was for sites 3, 4, 10, 16, 18, 24, 26 and 28 (Table 1).

The NRM of the remaining samples was usually fairly stable during thermal demagnetization (Figs. 2 and 3). Some samples, however, did not retain > 40% of the NRM after 150 °C (Fig. 4). Despite the low intensity, the mean directions are well (as for site 2) or reasonably well defined. In the latter case, the removal of an overprint component is obvious (Fig. 5).

Site-mean directions are based on the NRM cleaned at the highest possible step, AF or thermal.

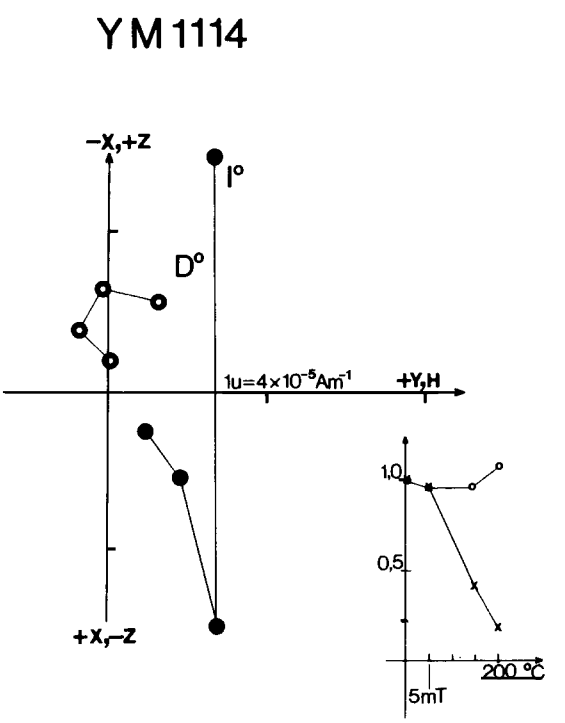


Fig. 4. Typical behaviour of Cretaceous samples during demagnetization. Krk, limestone, late Cretaceous, site 2. Details as in Fig. 2.

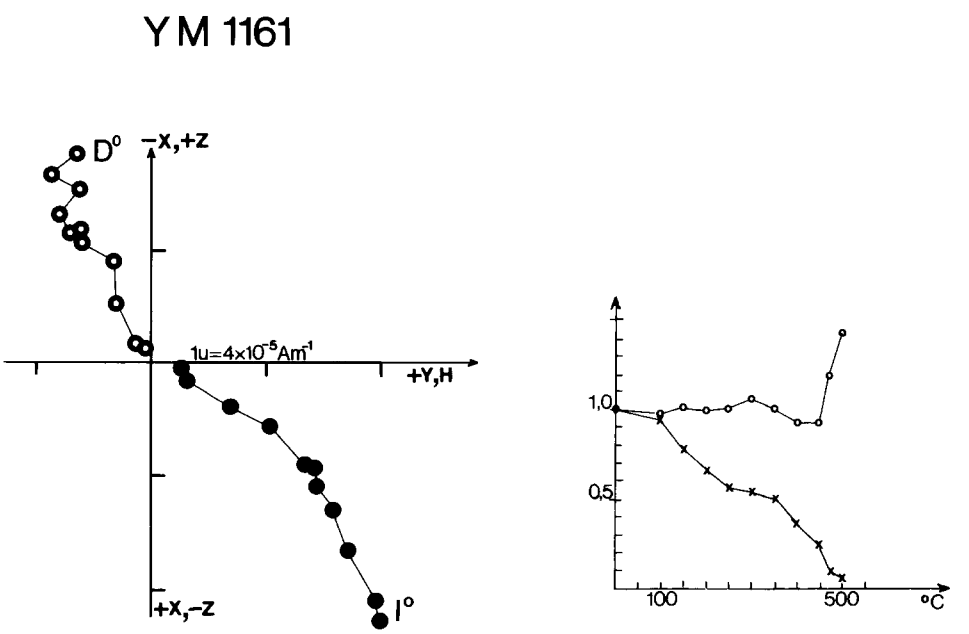


Fig. 3. Typical behaviour of Cretaceous samples during demagnetization. Krk, limestone early Cretaceous, site 8. Details as in Fig. 2.

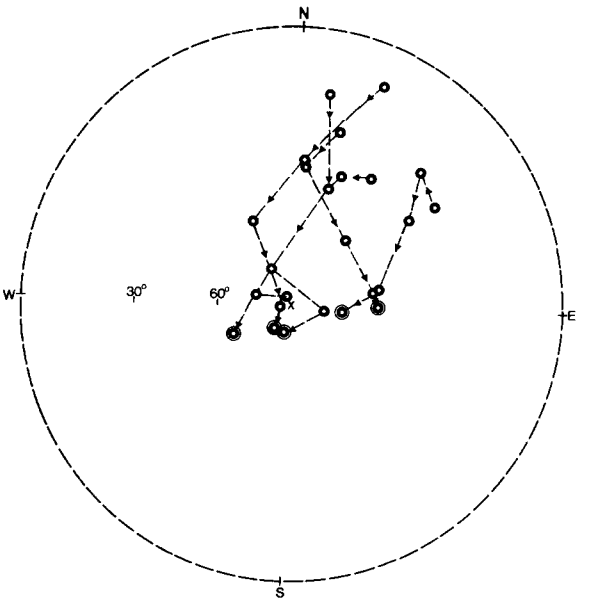


Fig. 5. Krk island, Turonian limestone, site 31. Change in direction of NRM during combined AF and thermal demagnetization (NRM to maximum of 10 mT AF and 200 °C thermal). Stereographic plot. All vectors point upward. Double symbols mark the end point of the cleaning.

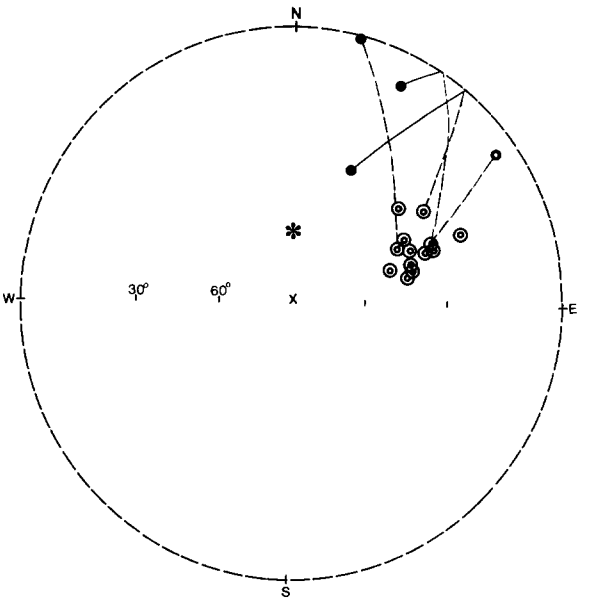


Fig. 7. Rab island, Eocene Foraminifera limestone, site 22–23. Change in direction of NRM during thermal cleaning. Stereographic projection. Stable end points shown in double circles were isolated above 300 °C. All vectors point upward.

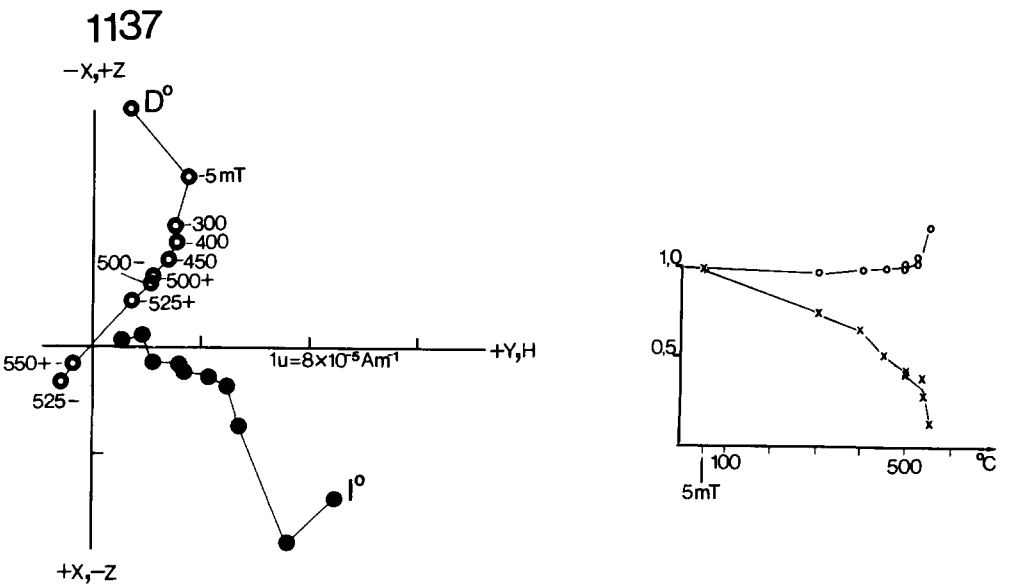


Fig. 6. Typical behaviour of Eocene Foraminifera limestone during demagnetization. Krk island. Site 5. Left: modified Zijderveld projection (circle: declination; dot: inclination) of the magnetic vector. Right: normalized (cross) intensity and susceptibility (circle) as a function of demagnetizing field and temperature. To estimate a possible error in the measured NRM due to a small rest field in the cooling chamber (less than 1 nT) after an increase of susceptibility, heating was repeated at the same temperature with x- and z-axes reversed (+ and − signs on the orthogonal plot). The change in polarity but not in direction above 525 °C should be noted.

Of the Eocene samples, Foraminifera limestone and flysch exhibit different behaviour. The removal of the overprint along the present field—if observable at all—requires 300 or 400 °C in the case of the Foraminifera limestones. As a rule, susceptibility remains unchanged up to 500 °C (Fig. 6). When the overprint is removed, ChRM with excellent values is defined (Table 1). The ChRM may be of one polarity (Fig. 7) or mixed (Figs. 8 and 9). In the latter case, the perfection of the cleaning is demonstrated by the antiparallelism of the vectors of different polarity (Fig. 8) or the antiparallelism of the magnetic directions before and after a polarity change has been observed for a whole site on heating (Fig. 9).

Although the NRM of the flysch seems to be less complex than that of the limestone (Fig. 10), satisfactory grouping of the magnetic vectors was not achieved, except for site 34 in Krk island and site 29 in Rab island. A great circle distribution was observed for sites 13, 15, 19, 20 and 21 from Pag island (Fig. 11). After correcting for local tectonics, a set of the intersections of the great

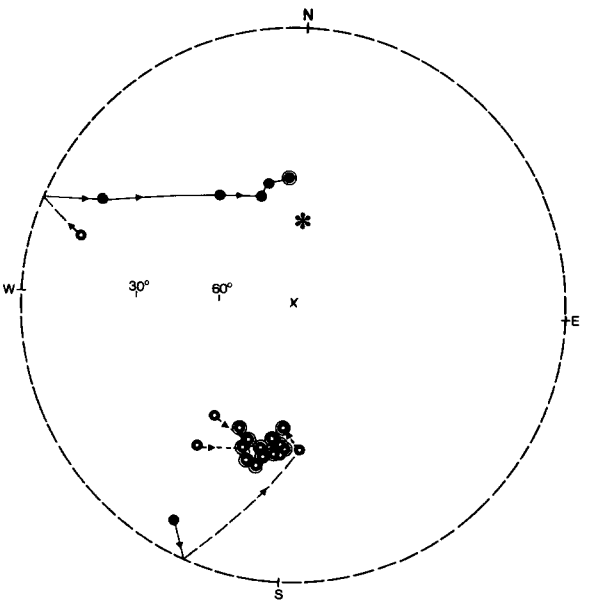


Fig. 8. Rab island, Eocene Foraminifera limestone, site 27. Change in direction of NRM during thermal cleaning. Stereographic projection. Stable end points are shown in double symbols. Vectors pointing upward are represented by circles, those pointing downward by dots.

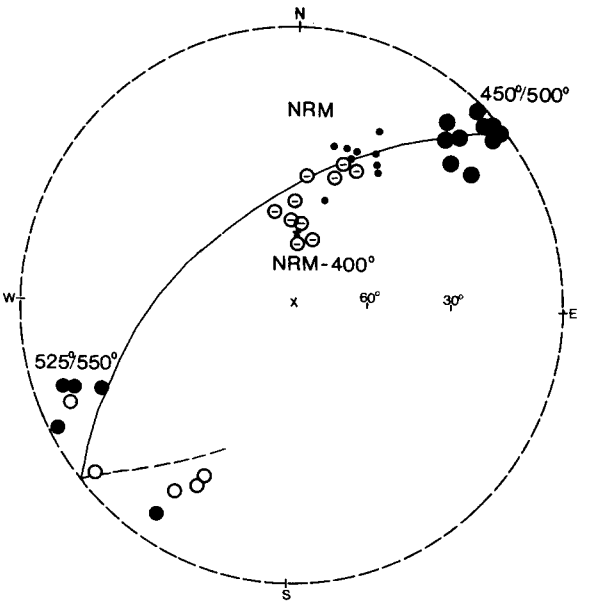


Fig. 9. Krk island, late Eocene Foraminifera limestone, site 5. Change in direction of NRM during heating. Stereographic projection. In the figure the direction of the NRM before cleaning is shown by small dots (positive inclinations), subtracted vectors (400 °C − initial NRM) in negative signs encircled (positive inclinations); the NRM isolated between 400 and 500 °C in larger dots (positive inclinations) are in the eastern hemisphere, whereas those observed after the increase in susceptibility, above 500 °C, are in the western hemisphere (dots positive, circles negative inclinations).

circles falls in the third quadrant of the lower hemisphere, thus suggesting that the pre-folding ChRM for the whole flysch basin of the present Pag island could have been rotated anticlockwise.

5. Interpretation of the paleomagnetic observations

From the Kvarner islands, five early (two from Cres, three from Krk) and 10 late Cretaceous sites (four from Cres, five from Krk, one from Pag) yielded statistically acceptable ChRM. The stability in time of the ChRM is indicated by the directions in the field system (Fig. 12a), for most of them depart significantly from the direction of the Earth's present field at the sampling area. On tilt correction, the early Cretaceous site means

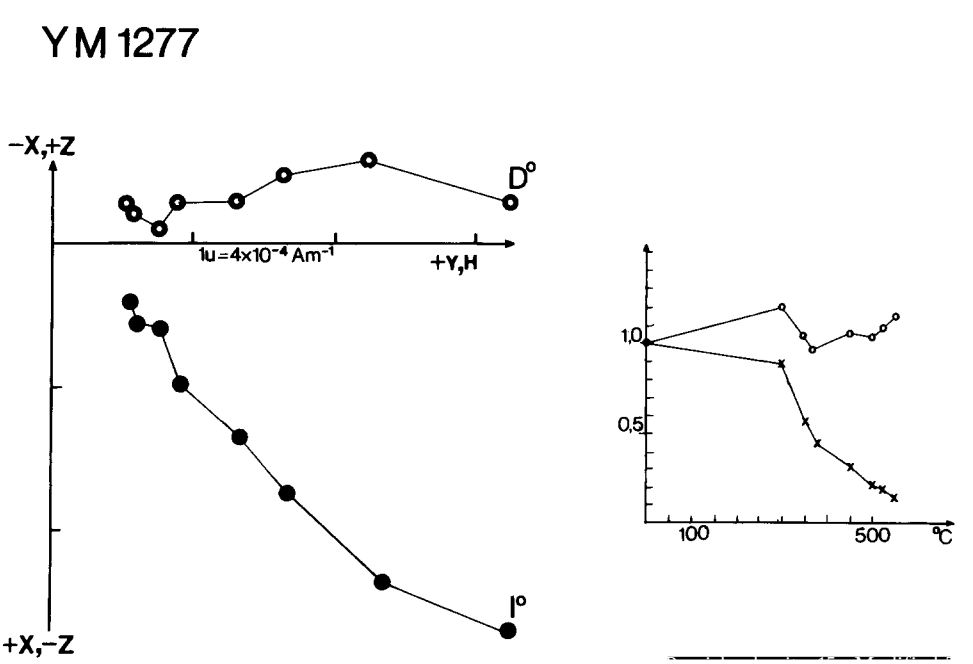


Fig. 10. Typical behaviour of Eocene flysch during thermal cleaning. Pag island, site 15. Modified Zijderveld projection (circle: declination; dot: inclination) of the magnetic vector (left) and normalized intensity (cross) and susceptibility (dots) as a function of temperature. Cleaning steps on the orthogonal plot correspond to those shown by the diagram on the right.

cluster, whereas the scatter for the late Cretaceous remains essentially unchanged (Fig. 12).

Although the values improve, the overall mean for the early Cretaceous shifts only slightly on tilt correction ($D = 343°$, $I = 41°$, $k = 8$ and $\alpha_{95} = 28°$ before, and $D_c = 350°$, $I_c = 45$, $k = 35$ and $\alpha_{95} = 13°$ after; five sites). For the late Cretaceous, the overall mean moves after correction ($D = 319°$, $I = 49°$, $k = 18$ and $\alpha_{95} = 12°$ before, $D_c = 333°$, $I_c = 50°$, $k = 15$ and $\alpha_{95} = 13°$ after; 10 sites) to the position found for the early Cretaceous.

The insignificant difference in the early and late Cretaceous overall means and the limited time interval represented by the sites (basically from Barremian–Aptian to Cenomanian–Turonian) permit the combination of 15 Cretaceous sites; thus we obtain $D = 327°$, $I = 47°$, $k = 12$ and $\alpha_{95} = 11°$ before and $D_c = 339°$, $I_c = 48°$, $k = 18$ and $\alpha_{95} = 9°$ after correction.

By comparing early and late Cretaceous paleomagnetic poles from Cres and Krk islands, respectively, with those determined for the fold belt north of 'autochthonous' Istria (Márton and Veljovic, 1987) and for the Trieste Karst (Márton and Pugliese, 1990) a better grouping of the paleomagnetic poles is found after (longitude = 245° E, latitude = 65° N, $k = 75$, $\alpha_{95} = 6°$), than before (longitude = 255° E, latitude = 54° N, $k = 12$, $\alpha_{95} = 17°$) tilt correction (Fig. 13). The improvement is significant, thus proving the pre-folding age of the magnetization (Márton and Pugliese, 1990) in a narrow zone of folds of Dinaric trend (Fig. 14).

According to the general practice in paleomagnetism, the paleomagnetic pole defined by 65° N, 245° E may be accepted as a Cretaceous pole. This pole is close to coeval poles known for the Southern Alps (Lowrie et al., 1980; Vanden-Berg and Wonders, 1980) and for Africa, but indicates clockwise rotation with respect to 'autochthonous' Istria (Márton and Veljovic, 1983) and —as expected for a tectonic unit of Southern Tethyan origin—anticlockwise rotation relative to stable Europe. The apparent clockwise rotation of a belt of such length with respect to the more rigid

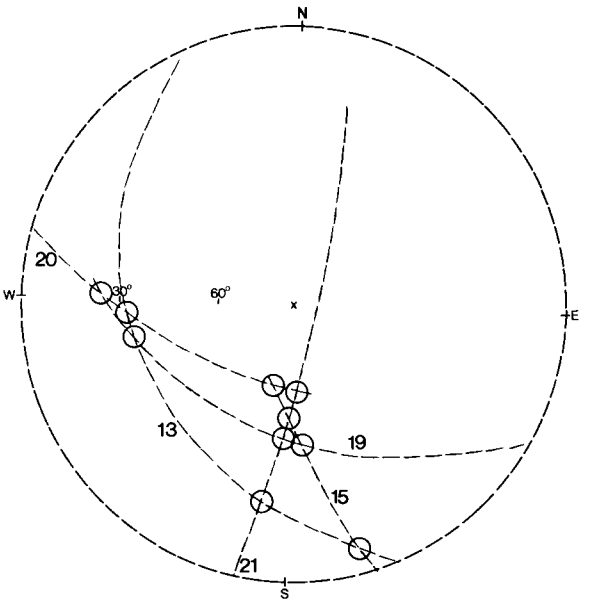


Fig. 11. Pag, Eocene flysch. Great circles joining the stable NRMs (tilt corrected) for each of sites 13, 15, 19, 20 and 21. The intersections on the lower hemisphere are encircled. The paleomagnetic directions of the flysch before folding may be sought in the area defined by the points of intersection, i.e. they must have positive inclination combined with a declination between 180 and 270°.

Adriatic area is not easily explained. Moreover, the belt probably continues to the west as far as the Southern Alps (Fig. 14, SA), which are bordered both on the north and the south by units which show excess counter-clockwise rotation with respect to Africa since the late Cretaceous (Eastern Alps, south of the Northern Calcareous Alps: Mauritsch and Becke, 1987; Umbrian Apennines: VandenBerg and Wonders, 1976).

It is unfortunate for the tectonic interpretation that the paleomagnetic study of the Eocene did not yield consistent directions. As Fig. 15 shows, both clockwise and counter-clockwise rotations occur in the Kvarner islands. In fact there is only one unit, the Omisalj-Baska syncline, where both senses were observed (sites 5 and 35 clockwise, site 33 counter-clockwise rotation). Otherwise, counter-clockwise rotation is prevalent both in the Foraminifera limestones (sites 22–23, 25, 27 and 33) and in the clastic members of the flysch (sites 29 and 34 and sites in Fig. 11). The amount of counter-clockwise rotation seems to vary considerably, even for sites with shallow dip, where the declination error due to imperfect tectonic correction is negligible. The ChRMs for the flysch complex are not uniform in inclination either, therefore they should be treated with caution. Despite allowance for an imprecision not expressed in the statistical parameters, we still think that at least a part of the variation in the declinations must be an expression of the chaotic movement of the flysch as opposed to the uniform response of the more consolidated Cretaceous to the same forces. Alternatively, some of the sampled Eocene sites, particularly those which show clockwise rotation, may not belong to the Dalmatian zone, but to the High Karst, as overlap of the Karst on the Dalmatian zone is assumed on the islands of Krk, Rab

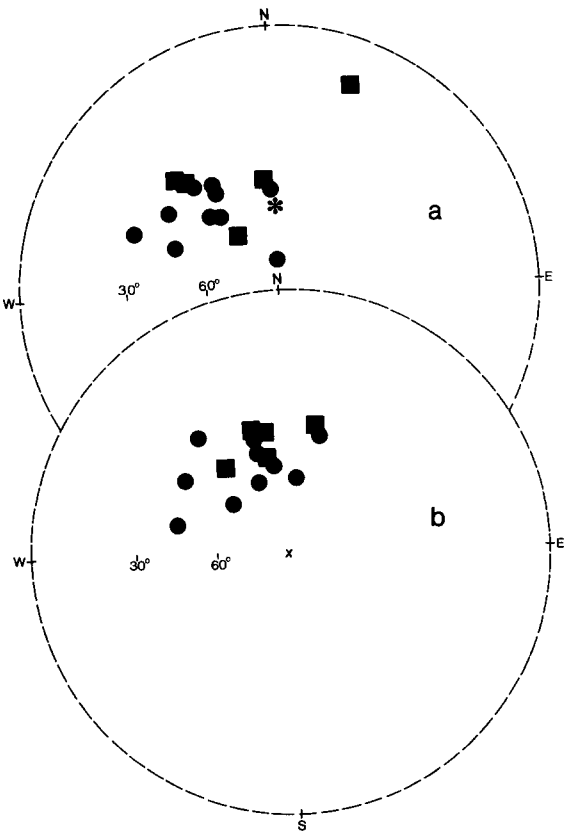


Fig. 12. Kvarner islands, Cretaceous site mean directions before (a) and after (b) tilt correction. Stereographic plot; all vectors point downward. Squares represent early Cretaceous, dots late Cretaceous. The asterisk marks the direction of the Earth's present field in the sampling area.

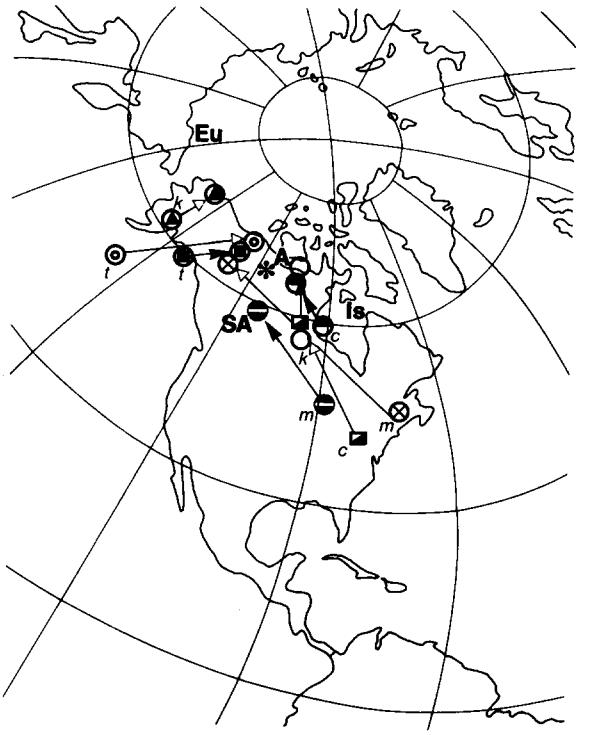


Fig. 13. Cretaceous paleomagnetic poles for the Kvarner islands (c: Cres; k: Krk), the para-autochthonous Dinaric north of 'autochthonous' Istria (m), the Trieste Karst (t). Poles computed for each area from site means before tilt correction are joined to the corresponding pole after correction by arrows (solid arrowheads: late Cretaceous poles; hollow arrowheads: early Cretaceous poles). The average Cretaceous pole (tilt corrected) for the fold belt ranging from the Trieste Karst to the Kvarner islands is shown by an asterisk. Cretaceous paleomagnetic poles for stable Europe (Eu), Africa (A), the Southern Alps (SA) and Istria (Is) are plotted for comparison.

and Pag (Celet, 1977). Whatever may be the explanation, the interpretation of the Cretaceous results is not aided.

The lack of post-Eocene sediments in the Dalmatian–Herzegovinian zone is another circumstance that inhibits the precise timing of the counterclockwise rotation observed for the Cretaceous.

In considering the relationship between the rigid part of the Adriatic area and the fold belt from Pag island to the Southern Alps, it is necessary to recall that we have no direct constraint on the timing of the post-Cretaceous rotation of Istria and Gargano.

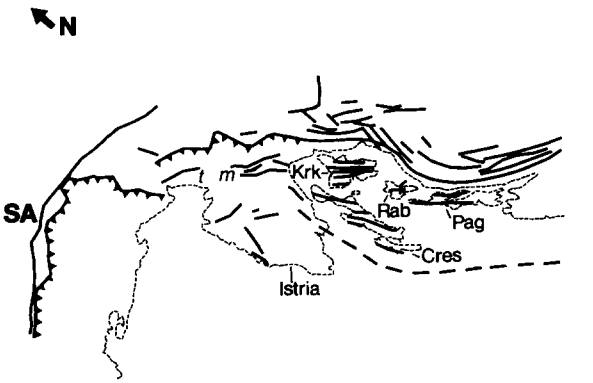


Fig. 14. Main trends of synclines, anticlines and major overthrusts in the northwestern part of the Periadriatic area. Sampling areas for which the poles of Fig. 13 are applicable are lettered as in Fig. 13.

Relying on the generally accepted model, namely that the Periadriatic area is the deformed margin of the 'Adriatic plate', we may envisage two models. One model assumes that both the margin and the rigid plate participated in the counter-clockwise rotation with respect to Africa and later the northwestern part of the

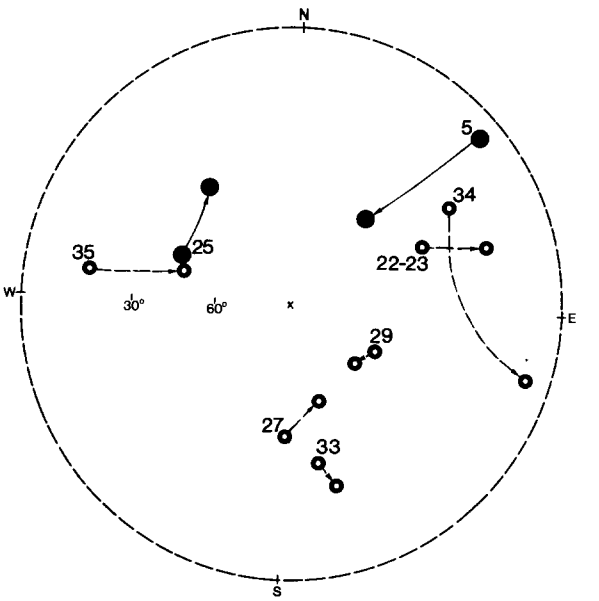


Fig. 15. Kvarner islands. Eocene site mean directions before and after tilt correction. Codes are the same as in Table 1. Stereographic projection. Circles: negative; dots: positive inclinations.

Dalmatian–Herzegovinian zone plus Trieste Karst (and Southern Alps?) rotated back in the clockwise sense, The most likely mechanism for the latter rotation is a kind of ball-bearing movement. Although observations of slide movements and translation parallel to the major Dinaric structures exist (e.g. Aubouin, 1963), the sense of the movement requires counter-clockwise rotation (advance and piling up of units from SE to NW).

In the second model a push from the SE that resulted in an advance of nappes from SE to NW is supposed to induce the excess counter-clockwise rotation of Istria and Gargano with respect to Africa. In this case, however, it is difficult to imagine why the rotation of the Dalmatic zone is retarded by about 30° with respect to that of the more rigid areas.

As both models have weak points, we must consider a third possibility, namely, that the angular difference in the counter-clockwise rotation between the rigid and mobile part of the 'Adriatic plate' predates the final rotation with respect to Africa. This tacitly implies that the deformed margin and the 'plate' itself were welded together as late as the Miocene, and then both participated in a final counter-clockwise rotation.

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